MODULE *Agent*

EXTENDS *Naturals*

CONSTANT *MaxSteps*

VARIABLES *state*, *steps*

$$
\begin{aligned}
&\mathit{Init} \triangleq \\
&\quad \land\ \mathit{state} = \text{“reasoning”} \\
&\quad \land\ \mathit{steps} = 0
\end{aligned}
$$

$$
\begin{aligned}
&\mathit{Reason} \triangleq \\
&\quad \land\ \mathit{state} = \text{“reasoning”} \\
&\quad \land\ \mathit{steps} < \mathit{MaxSteps} \\
&\quad \land\ \mathit{state}' \in \{\text{“acting”},\ \text{“returning”}\} \\
&\quad \land\ \mathit{steps}' = \mathit{steps} + 1
\end{aligned}
$$

$$
\begin{aligned}
&\mathit{Act} \triangleq \\
&\quad \land\ \mathit{state} = \text{“acting”} \\
&\quad \land\ \mathit{state}' = \text{“reasoning”} \\
&\quad \land\ \text{UNCHANGED } \mathit{steps}
\end{aligned}
$$

$$
\begin{aligned}
&\mathit{Return} \triangleq \\
&\quad \land\ \mathit{state} = \text{“returning”} \\
&\quad \land\ \mathit{state}' = \text{“done”} \\
&\quad \land\ \text{UNCHANGED } \mathit{steps}
\end{aligned}
$$

$$\mathit{Next} \triangleq \mathit{Reason} \lor \mathit{Act} \lor \mathit{Return}$$

$$\mathit{Spec} \triangleq \mathit{Init} \land \Box[\mathit{Next}]_{\langle \mathit{state},\, \mathit{steps} \rangle}$$

$$\mathit{Terminates} \triangleq \Diamond(\mathit{state} = \text{“done”})$$

$$\mathit{Bounded} \triangleq \Box(\mathit{steps} \leq \mathit{MaxSteps})$$